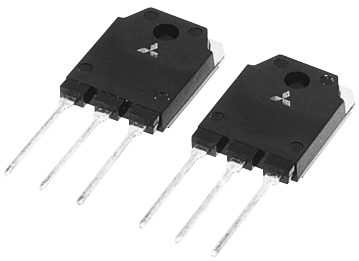
PRELIMINARY
Notice: This is not a final specification.
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MITSUBISHI Pch POWER MOSFET

FX50SMJ-2

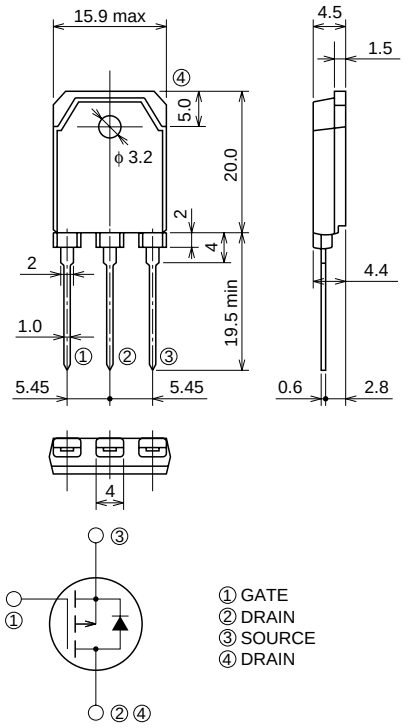
HIGH-SPEED SWITCHING USE

FX50SMJ-2



- 4V DRIVE
- V_{DSS} ..... –100V
- $r_{DS\ (ON)\ (MAX)}$ 50mΩ
- I_D ..... –50A
- Integrated Fast Recovery Diode (TYP.) 100ns

OUTLINE DRAWING Dimensions in mm



TO-3P

APPLICATION

Motor control, Lamp control, Solenoid control
DC-DC converter, etc.

MAXIMUM RATINGS (T_c = 25°C)

Symbol	Parameter	Conditions	Ratings	Unit
V_{DSS}	Drain-source voltage	V_{GS} = 0V	–100	V
V_{GSS}	Gate-source voltage	V_{DS} = 0V	±20	V
I_D	Drain current		–50	A
I_{DM}	Drain current (Pulsed)		–200	A
I_{DA}	Avalanche drain current (Pulsed)	L = 30μH	–50	A
I_S	Source current		–50	A
I_{SM}	Source current (Pulsed)		–200	A
P_D	Maximum power dissipation		150	W
T_{ch}	Channel temperature		–55 ~ +150	°C
T_{stg}	Storage temperature		–55 ~ +150	°C
—	Weight	Typical value	4.8	g

MITSUBISHI ELECTRIC

PRELIMINARY
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FX50SMJ-2

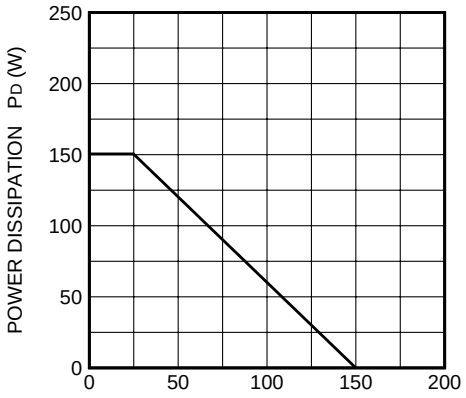
HIGH-SPEED SWITCHING USE

ELECTRICAL CHARACTERISTICS (Tch = 25°C)

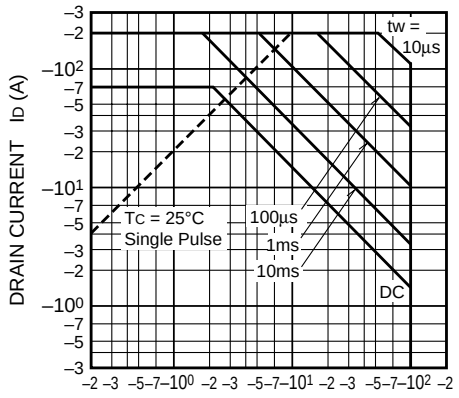
Symbol	Parameter	Test conditions	Limits			Unit
			Min.	Typ.	Max.	
$V_{(BR)DSS}$	Drain-source breakdown voltage	I_D = –1mA, V_{GS} = 0V	–100	—	—	V
I_{GSS}	Gate-source leakage current	V_{GS} = ±20V, V_{DS} = 0V	—	—	±0.1	µA
I_{DSS}	Drain-source leakage current	V_{DS} = –100V, V_{GS} = 0V	—	—	–0.1	mA
$V_{GS(th)}$	Gate-source threshold voltage	I_D = –1mA, V_{DS} = –10V	–1.0	–1.5	–2.0	V
$r_{DS(ON)}$	Drain-source on-state resistance	I_D = –25A, V_{GS} = –10V	—	39	50	mΩ
$r_{DS(ON)}$	Drain-source on-state resistance	I_D = –25A, V_{GS} = –4V	—	47	61	mΩ
$V_{DS(ON)}$	Drain-source on-state voltage	I_D = –25A, V_{GS} = –10V	—	–0.98	–1.25	V
$\lvert y_{fs} \rvert$	Forward transfer admittance	I_D = –25A, V_{DS} = –10V	—	49.2	—	S
C_{iss}	Input capacitance	V_{DS} = –10V, V_{GS} = 0V, f = 1MHz	—	11130	—	pF
C_{oss}	Output capacitance		—	896	—	pF
C_{rss}	Reverse transfer capacitance		—	480	—	pF
$t_{d(on)}$	Turn-on delay time	V_{DD} = –50V, I_D = –25A, V_{GS} = –10V, R_{GEN} = R_{GS} = 50Ω	—	57	—	ns
t_r	Rise time		—	118	—	ns
$t_{d(off)}$	Turn-off delay time		—	828	—	ns
t_f	Fall time		—	380	—	ns
V_{SD}	Source-drain voltage	I_S = –25A, V_{GS} = 0V	—	–1.0	–1.5	V
$R_{th(ch-c)}$	Thermal resistance	Channel to case	—	—	0.83	°C/W
t_{rr}	Reverse recovery time	I_S = –50A, dis/dt = 100A/µs	—	100	—	ns

PERFORMANCE CURVES

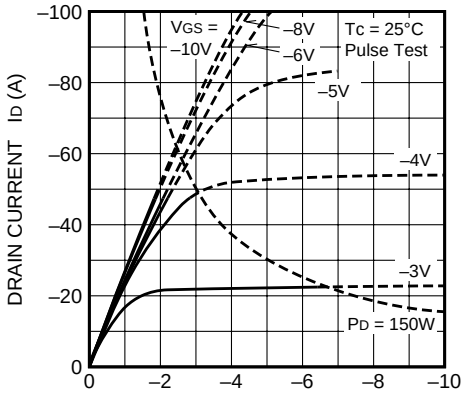
POWER DISSIPATION DERATING CURVE



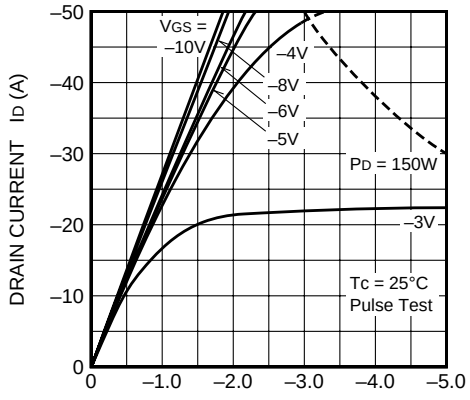
MAXIMUM SAFE OPERATING AREA



OUTPUT CHARACTERISTICS (TYPICAL)



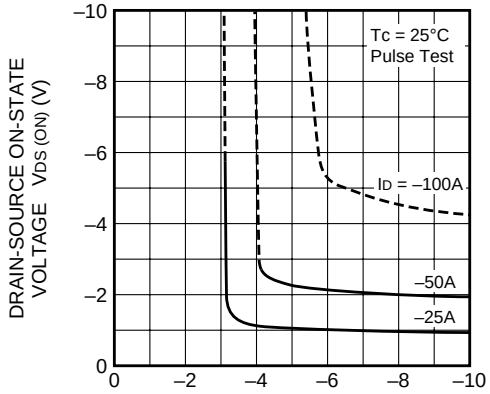
OUTPUT CHARACTERISTICS (TYPICAL)



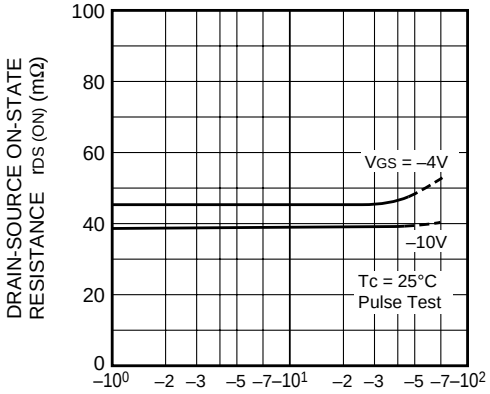
MITSUBISHI ELECTRIC

PRELIMINARY
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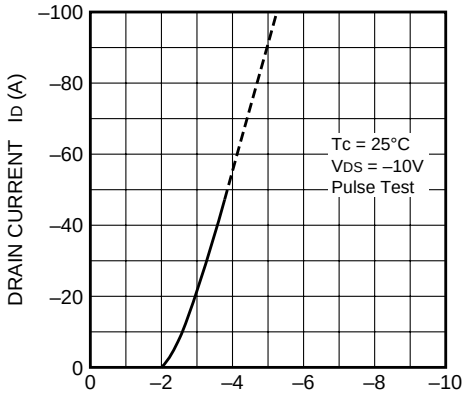
ON-STATE VOLTAGE VS. GATE-SOURCE VOLTAGE (TYPICAL)



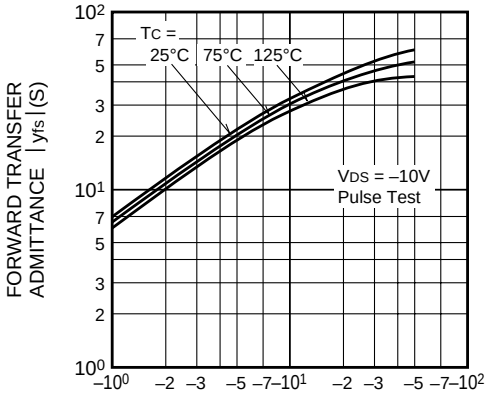
ON-STATE RESISTANCE VS. DRAIN CURRENT (TYPICAL)



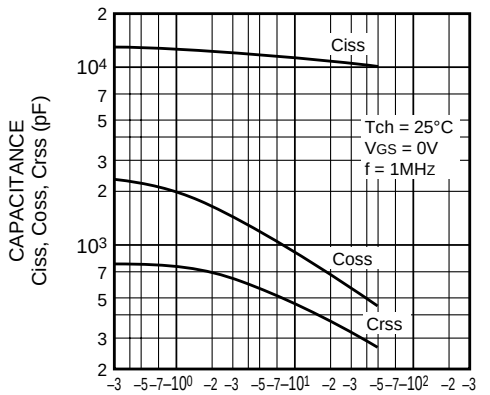
TRANSFER CHARACTERISTICS (TYPICAL)



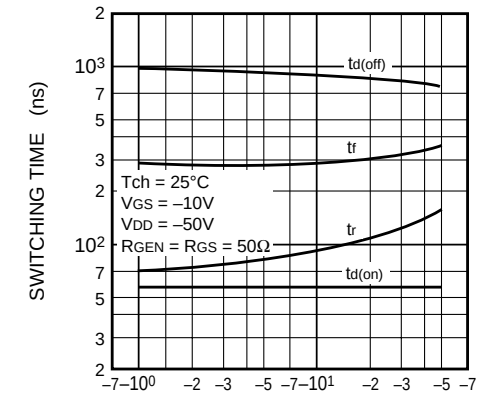
FORWARD TRANSFER ADMITTANCE VS.DRAIN CURRENT (TYPICAL)



CAPACITANCE VS. DRAIN-SOURCE VOLTAGE (TYPICAL)

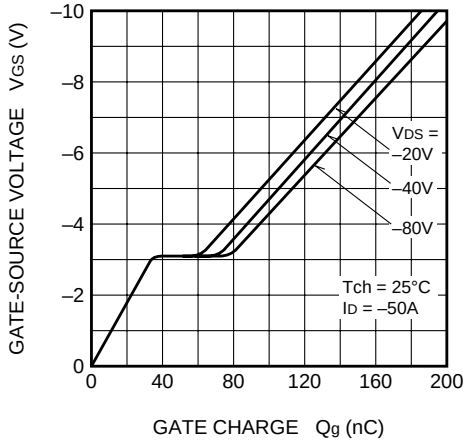


SWITCHING CHARACTERISTICS (TYPICAL)

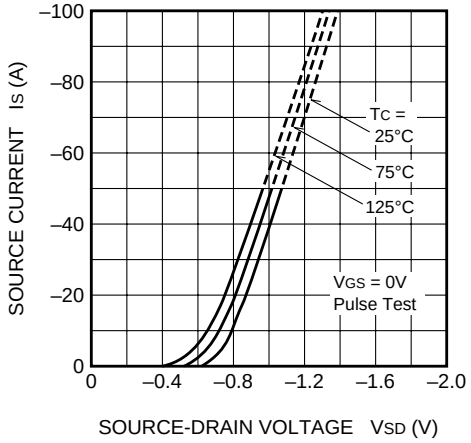


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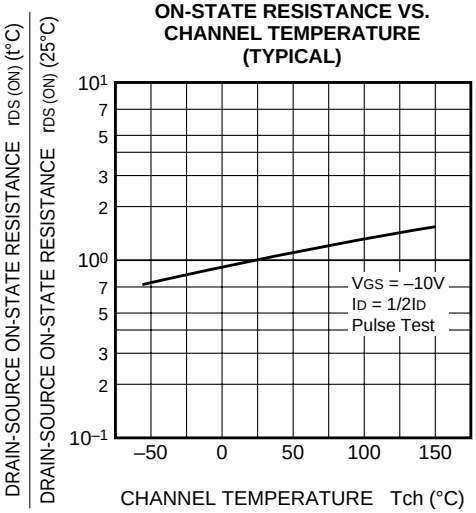
GATE-SOURCE VOLTAGE VS.GATE CHARGE (TYPICAL)



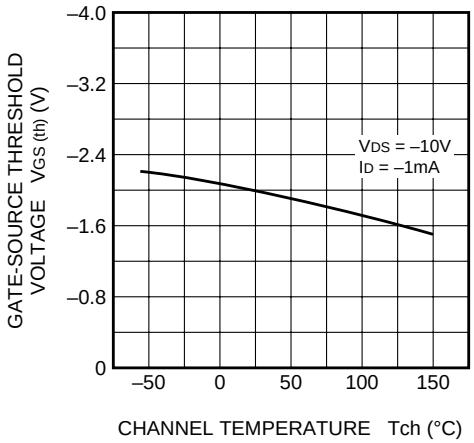
SOURCE-DRAIN DIODE FORWARD CHARACTERISTICS (TYPICAL)



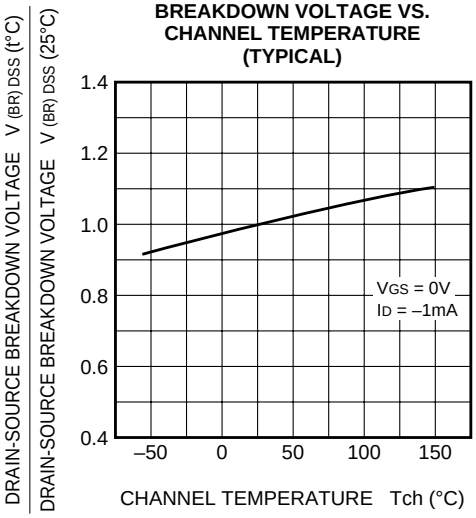
ON-STATE RESISTANCE VS. CHANNEL TEMPERATURE (TYPICAL)



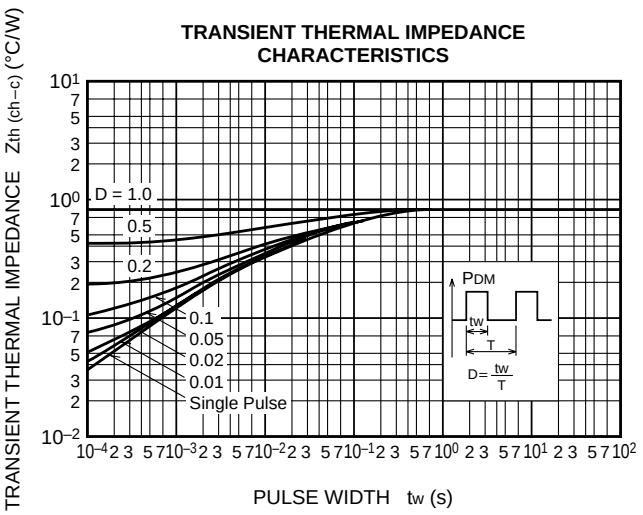
THRESHOLD VOLTAGE VS. CHANNEL TEMPERATURE (TYPICAL)



BREAKDOWN VOLTAGE VS. CHANNEL TEMPERATURE (TYPICAL)



TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS



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